

Mathematics Notes

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Exercise 1. Prove that a polynomial of degree n is uniformly continuous on $\mathbb{R}$ if and only if $n = 0$ or 1 .

Proof. Let f be a polynomial of degree n and suppose that $n \geq 2$. We wish to prove that for any $\delta > 0$, there exist points $x, y \in \mathbb{R}$ such that $|x - y| < \delta$ and $|f(x) - f(y)| \geq 1$. Notice that for all $\delta > 0$, there exists $M \in \mathbb{R}$ such that $x > M$ implies $|f'(x)| > 2/\delta$, since $|f'(x)| \rightarrow \infty$ as $x \rightarrow \infty$. Let $x = M$ and $y = M + \delta/2$. By the Mean Value Theorem, choose $\xi \in (x, y)$ such that

$$f(x) - f(y) = f'(\xi)(x - y).$$

Since $\xi > M$ and $|x - y| = \delta/2$, we have

$$\begin{aligned} |f(x) - f(y)| &= |f'(\xi)| |x - y| \\ &> (2/\delta) \cdot |x - y| = 1. \end{aligned}$$

Therefore, f is not uniformly continuous on $\mathbb{R}$ if $n \geq 2$. Suppose that $n = 1$. We write $f(x) = ax + b$ with $a \neq 0$ and take $\delta = \varepsilon/|a|$ for every $\varepsilon > 0$. Then $|x - y| < \delta$ and $x, y \in \mathbb{R}$ imply $|f(x) - f(y)| = |a||x - y| < |a|\delta = \varepsilon$. It is obvious that f is uniformly continuous on $\mathbb{R}$ if $n = 0$. We conclude that f is uniformly continuous on $\mathbb{R}$ if and only if $n = 0$ or 1 . $\square$